WORLDPRISM: 3D Consistency for Video World Models via Bidirectional Cross-Space Verification

Anonymous Author(s)
Affiliation
Address
email

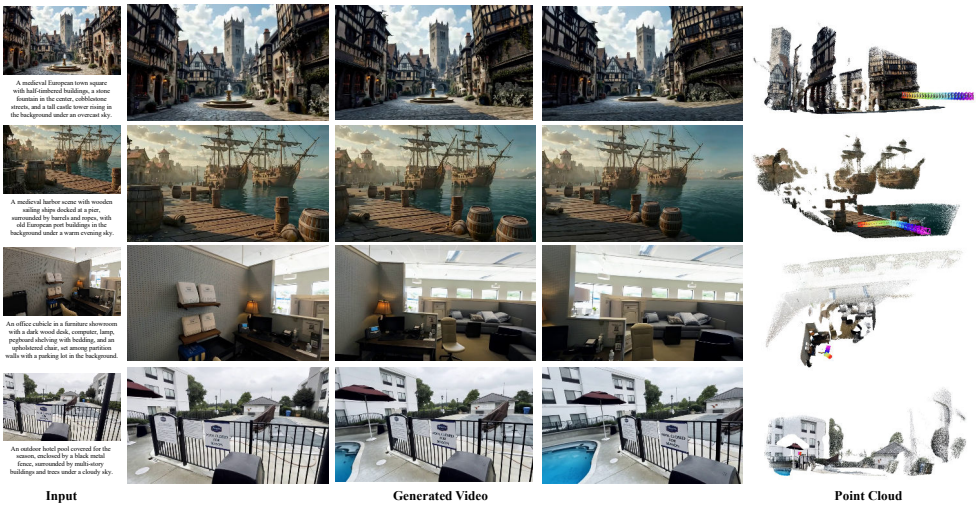


Figure 1: WORLDPRISM enables geometrically consistent world modeling for video generation via reinforcement learning. Given reference images, text prompts, and action sequences, it produces high-quality videos with strong 3D consistency. We visualize both the generated frames and their corresponding 3D reconstructions.

Abstract

Video world models aim to generate temporally coherent visual observations conditioned on actions, but often suffer from 3D geometric inconsistencies such as spatial drift, object deformation, and perspective violations, which limit their reliability in downstream tasks. We propose WORLDPRISM, a post-training framework that improves 3D consistency via reinforcement learning without modifying the base model architecture. The central challenge lies in designing a reliable reward signal. To address this, we introduce *Bidirectional Geometry-Flow Verification*, a bidirectional cross-space mechanism that evaluates consistency through two complementary paths: (i) lifting 2D optical flow into 3D to measure position-space agreement, and (ii) projecting reconstructed 3D geometry to match the point correspondence. By leveraging these two independent and complementary signals, this design mitigates the blind spots of unilateral 2D evaluation. We further incorporate hierarchical temporal evaluation to capture both local and long-range geometric consistency. Experiments on diverse datasets demonstrate that WORLDPRISM achieves state-of-the-art 3D consistency while preserving visual quality and generative diversity.

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1 Introduction

Video world models [1, 2, 3, 4, 5, 6] learn to autoregressively generate visual observations conditioned on actions, functioning as learned simulators that support downstream tasks such as planning, navigation, and manipulation. A critical quality axis for such models is *3D geometric consistency*, the requirement that a generated video faithfully depicts a single, coherent 3D world [7, 8, 9, 10]. Yet even models pretrained on internet-scale data still exhibit object deformation, spatial drift, and perspective violations under camera motion [11, 12], directly undermining downstream decision-making: an embodied agent relying on spatially inconsistent predictions can produce erroneous plans.

Various strategies have been explored to improve this consistency. One line of work maintains explicit 3D memory during generation, storing per-frame scene representations [13, 14], accumulated point clouds [15], or pose-aware latent memory banks [16]. Camera-conditioned approaches [17, 18, 19] take a different route, encoding explicit camera trajectories such as Plücker embeddings or extrinsic matrices as spatiotemporal control signals injected into the denoising process. Geometric regularization methods [10] instead introduce 3D losses during training to encourage structural fidelity. While effective to varying degrees, these approaches either modify the model architecture and inference pipeline with substantial computational overhead [13, 15] or require large-scale 3D ground-truth data that is difficult to curate [10, 17], limiting their applicability as a general-purpose alignment mechanism for pretrained video world models [11, 20].

These limitations motivate post-training as an attractive alternative: rather than modifying the model architecture or curating extensive 3D supervision, one can align a pretrained video model toward better geometric consistency using reinforcement learning [21, 22, 23, 24, 25]. By preserving the original generative capacity while steering its outputs, this paradigm shifts the core difficulty from architectural design to reward design: *how to define and measure 3D consistency as a reliable training signal?* Existing approaches [11, 12] adopt a *unilateral* evaluation strategy that assesses geometric fidelity through a single forward projection into 2D. For example, VideoGPA [11] reprojects 3D structure to the image plane and measures RGB differences, while Kupyn et al. [12] evaluate epipolar constraints via the Sampson error [26]. However, optimizing solely on such 2D-space rewards introduces inherent blind spots. First, the irreversible 3D-to-2D projection only reveals visible regions, potentially hiding underlying geometric inconsistencies. Second, evaluating in RGB space often entangles true structural errors with view-dependent appearance variations [27]. Driven by these incomplete signals, the model's optimization is limited by what can be observed, imposing a ceiling on the 3D consistency achievable through post-training.

To address this challenge, we introduce WORLDPRISM, a post-training framework with *Bidirectional Geometry-Flow Verification (BGFV)* that aims to enhance the 3D consistency of current video world models. Specifically, it includes two independent and complementary streams: *Flow-Guided Geometry Verification (F2G)* and *Geometry-Induced Flow Verification (G2F)*. Given a video sequence generated by video world models, we utilize off-the-shelf geometry foundation models [28, 29] and an optical flow estimation model [30] to estimate 3D geometry and 2D flow. In *F2G*, we first use the estimated optical flow to find matching points between the source and target frames, then project them into 3D space to compute the 3D position error. In *G2F*, we first reproject the reconstructed 3D point clouds from the source frame to the target frame to produce geometry-induced 2D flow, then supervise it using the directly estimated optical flow. By spanning both spaces with independent signals, each stream covers the structural blind spots of the other: geometric violations invisible under one stream are exposed by the complementary one, as shown in Fig. 2. Moreover, since the two streams rely on *independent* and *orthogonal* external models, their mutual agreement on high error constitutes compelling evidence of genuine 3D violations rather than artifacts of any single perceptual backbone. WORLDPRISM further organizes this reward into a hierarchical temporal evaluation spanning local, structural, and global scales with curriculum-scheduled weighting, and integrates the resulting signal into a DiffusionNFT [31]-based reinforcement learning pipeline for effective post-training.

We validate WORLDPRISM on HunyuanVideo [4, 32] equipped with the WorldPlay [3] autoregressive action conditioning, where it yields substantial improvements in 3D geometric consistency while faithfully preserving visual quality and generative diversity. Our contributions are three-fold:

- We identify a fundamental bottleneck in existing post-training methods for 3D-consistent video world models: they rely on a 2D projection of a single evaluation space to assess

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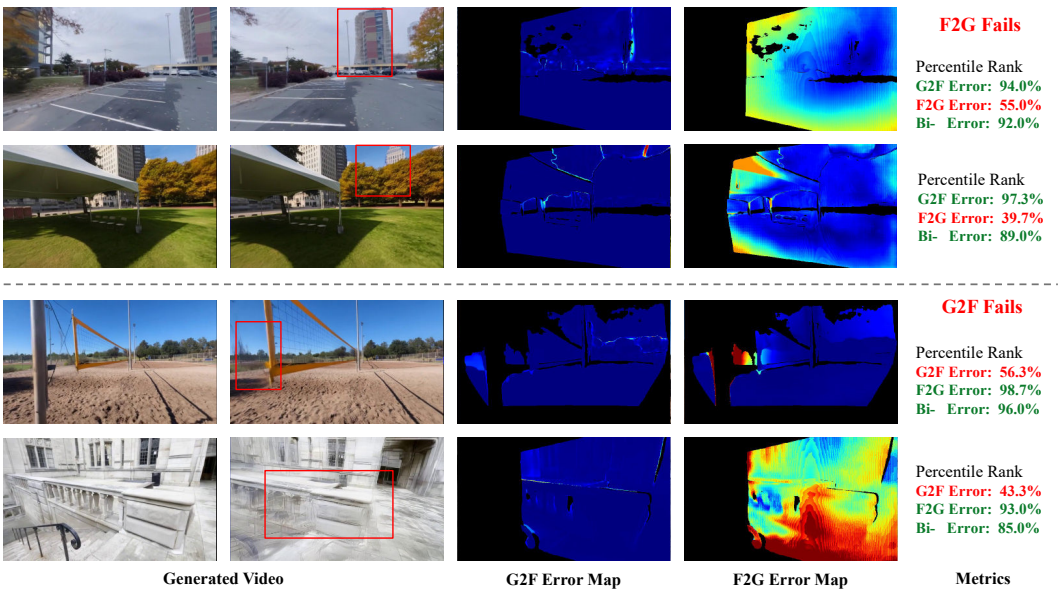


Figure 2: **Reward Robustness Case Study.** Failures in single-path verification are detected by our bidirectional reward. Percentile ranks of G2F and F2G errors show each frame pair's position among all processed pairs, where higher values indicate higher ranks.

consistency, thereby inheriting the systematic blind spots of 2D projections, and lack an independent self-correction mechanism.

- We design a *Bidirectional Geometry-Flow Verification* mechanism comprising two complementary streams that enable both Geometry-Induced Flow Verification and Flow-Guided Geometry Verification. The bidirectional design delivers a robust 3D consistency signal, minimizing inductive biases inherent in individual perception models.
- We propose WORLDPRISM, a complete post-training framework for video world models, and demonstrate that it can significantly improve 3D consistency on the HunyuanVideo model (using WorldPlay as the action model).

2 Related Work

3D Consistency in Video Generation. Approaches to improving the 3D geometric consistency of generated videos can be broadly categorized by *where* geometric constraints are imposed in the model lifecycle. *Architecture-level methods* integrate explicit 3D representations into the generation pipeline. DAS [14] introduces long-term 3D trajectories for controllable world modeling; Zhang et al. [13] maintain a persistent 3D scene updated via differentiable rendering; Wu et al. [15] accumulate a global point cloud as a geometric prior for frame synthesis; and WorldMem [16] employs a pose-indexed latent memory queried via cross-attention to retrieve prior geometry. *Camera-conditioned methods* [17, 18, 19] encode camera trajectories as explicit control signals. CameraCtrl [17] injects Plücker ray embeddings into temporal attention for 6-DoF control; CameraCtrl-II [18] extends this to dynamic scenes by disentangling camera and object motion; and Trajectory-Attention [19] introduces a trajectory-aware attention branch to complement temporal attention, enabling coordinated motion control and content generation. *Geometric regularization* [10] incorporates depth and epipolar consistency losses during training to improve structural fidelity without modifying the inference architecture. While effective, these approaches impose constraints *before or during* generation, either requiring architectural modifications with added complexity or relying on large-scale 3D supervision, making them difficult to apply to pretrained models. This motivates post-training as a complementary paradigm, aligning pretrained models *after* generation through reward-driven optimization.

Post-Training for Video World Models. Post-training through reinforcement learning has emerged as a flexible paradigm for aligning pretrained diffusion models without architectural modification.

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Early work applies RLHF-style optimization to image diffusion models for text alignment and aesthetic quality [22, 25, 21]. More recent methods extend this to flow matching video models: FlowGRPO [23] and DanceGRPO [24] adapt group relative policy optimization to the continuous denoising setting, while DiffusionNFT [31] introduces a negative-aware fine-tuning strategy that explicitly leverages low-reward samples as contrastive signals. WorldCompass [33] applies DiffusionNFT to video world models with action-following and visual quality rewards. As established in these works, the post-training paradigm shifts the core difficulty from architectural design to *reward design*, and the quality of the reward signal directly determines the ceiling of achievable alignment.

Reward Design for 3D Consistency. When post-training targets 3D consistency specifically, the central challenge becomes how to measure geometric fidelity reliably enough to serve as a training signal. Existing approaches predominantly adopt a *unilateral* evaluation strategy: they assess consistency through a single verification direction within a single evaluation space. WorldScore [7] applies DROID-SLAM [34] to compute dense reprojection error; VideoGPA [11] and GeCo [35] leverage geometry foundation models [36, 37] for depth-based reprojection; Kupyn et al. [12] evaluate epipolar constraints with Sampson error [26]; and VIGOR [38] computes cross-frame reprojection error against independently tracked correspondences. Despite differences in signal sources, these methods share a common structural limitation: committing to a single projection direction inherits the systematic blind spots of that particular mapping, and operating in a single evaluation space provides no independent mechanism to distinguish model noise from genuine geometric violations. This places an inherent ceiling on the 3D consistency achievable through such rewards. WORLDPRISM lifts this ceiling through two complementary paths operating in opposite directions, one lifting 2D correspondences into 3D space, the other projecting 3D geometry into 2D space, cross-check geometry-flow consistency in both directions. Because the paths rely on external models with orthogonal failure modes, their mutual agreement on high error constitutes compelling evidence of genuine 3D violations rather than artifacts of any single perceptual backbone.

3 Method: WORLDPRISM

3.1 Problem Formulation

Given an image condition c, a sequence of discrete actions $a_{1:N}$, and a target clip index n to optimize, we first autoregressively generate the preceding $n-1$ clips as shared prefix context $x_{1:n-1}$, then produce G candidate rollouts for the n-th clip:

$$x_{1:n-1} = \pi_\theta(\cdot \mid a_{1:n-1}, c), \quad x_n^{(i)} = \pi_\theta(\cdot \mid x_{1:n-1}, a_n, c), \quad i = 1, \ldots, G. \tag{1}$$

Each clip is generated via a flow matching diffusion transformer operating in latent space. Let $\mathbf{z}_0 = \mathcal{E}(x_n)$ denote the encoded latent of a target clip. The forward process constructs noisy latents via linear interpolation:

$$\mathbf{z}_\sigma = (1 - \sigma)\, \mathbf{z}_0 + \sigma\, \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{2}$$

where $\sigma \in [0, 1]$ follows a shifted cosine schedule. The model $v_\theta(\mathbf{z}_\sigma, \sigma, x_{1:n-1}, a_n, c)$ learns to predict the velocity field that denoises $\mathbf{z}_\sigma$ toward $\mathbf{z}_0$.

Each candidate is scored by a reward function $R(x_n^{(i)}, a_n, c)$ that quantifies the 3D geometric consistency of the generated clip. The central challenge is the design of R: a reward that reliably captures 3D violations without the systematic blind spots of unilateral 2D evaluation. We address this through Bidirectional Geometry-Flow Verification (§3.2), a bilateral cross-space reward that cross-checks optical flow against 3D reconstruction through two complementary verification paths.

3.2 Bidirectional Geometry-Flow Verification

Rather than committing to a single projection direction, we cross-check 3D consistency through two independently sourced geometric signals, dense optical flow $\mathbf{f}_{i \to j} = \mathrm{UFM}(\mathbf{I}_i, \mathbf{I}_j)$ from a pretrained flow model [30], and temporally aligned per-pixel 3D point maps $\{\mathbf{P}_t \in \mathbb{R}^{H \times W \times 3}\}$ from a pretrained reconstruction model [28]—across both position space and motion space. In a geometrically consistent video these two signals should agree after being expressed in a common coordinate frame; systematic disagreement constitutes evidence of genuine 3D violations rather than artifacts of any single perceptual model. The two verification paths are detailed below.

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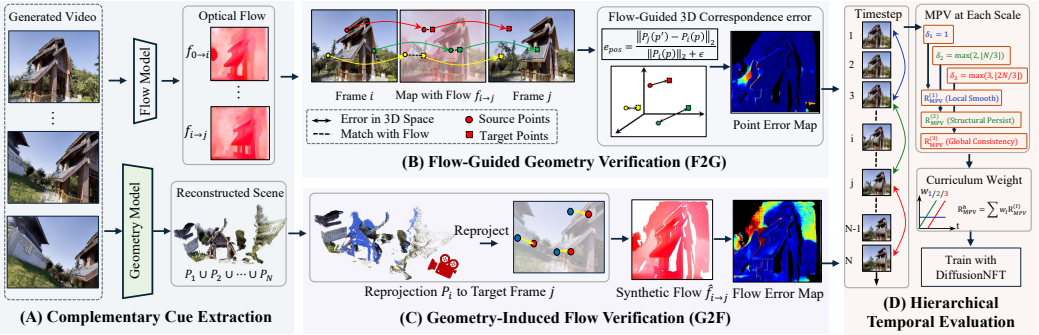


Figure 3: **Overview of WORLDPRISM**. Given a reference image, text prompt, and action sequence, the model autoregressively generates video clips using a flow-matching diffusion transformer. For each target clip, multiple candidates are sampled and evaluated by our proposed *Bidirectional Geometry-Flow Verification ((BGFV))* reward. BGFV performs bilateral cross-space verification : (i) a forward path lifts 2D optical flow into 3D to measure position-space consistency, and (ii) a reverse path projects reconstructed 3D geometry to synthesize flow for motion-space verification. The two signals are fused via a geometric mean and further aggregated through hierarchical temporal evaluation across multiple time offsets. The resulting reward guides reinforcement learning-based post-training to improve 3D geometric consistency without modifying the base model architecture.

3.2.1 Flow-Guided Geometry Verification (F2G, Forward Path)

The forward path lifts 2D correspondences into 3D and evaluates their agreement in *position space*. This path directly addresses a key blind spot of unilateral 2D evaluation: geometric violations that are invisible after projection, such as depth ordering errors or lateral drift at similar screen positions, become explicit once correspondences are examined in full 3D coordinates.

Concretely, given a frame pair $(\mathbf{I}_i, \mathbf{I}_j)$ with estimated flow $\mathbf{f}_{i\to j}$, each pixel $\mathbf{p}$ in frame i is mapped to its corresponding location $\mathbf{p}' = \mathbf{p} + \mathbf{f}_{i\to j}(\mathbf{p})$ in frame j. We then look up the reconstructed 3D positions at both ends: $\mathbf{P}_i(\mathbf{p})$ from the source frame and $\mathbf{P}_j(\mathbf{p}')$ from the target frame, with both positions expressed in the same reconstructed coordinate frame. In a geometrically consistent video, these two 3D positions should coincide.

Since monocular reconstruction can produce point maps with an arbitrary per-pixel scale factor, direct Euclidean comparison is unreliable. We therefore adopt a scale-invariant relative distance:

$$e_{\text{pos}}(\mathbf{p}) = \frac{\|\mathbf{P}_j(\mathbf{p}') - \mathbf{P}_i(\mathbf{p})\|_2}{\|\mathbf{P}_i(\mathbf{p})\|_2 + \epsilon}. \tag{3}$$

This metric is inherently scale-invariant: for any positive scaling factor k, $\|k\mathbf{P}_j - k\mathbf{P}_i\|/\|k\mathbf{P}_i\| = \|\mathbf{P}_j - \mathbf{P}_i\|/\|\mathbf{P}_i\|$, naturally reducing the per-frame scale ambiguity of monocular reconstruction. Unlike depth-only metrics that measure only the z-component, this formulation captures lateral drift across the full XYZ space, providing a more complete position-space constraint. The path-level error averages over all valid correspondences: $E_{\text{pos}} = \frac{1}{|\Omega_{\text{pos}}|}\sum_{\mathbf{p}\in\Omega_{\text{pos}}} e_{\text{pos}}(\mathbf{p})$, where Ω_{pos} contains pixels with in-bound flow correspondences and valid reconstructed 3D points.

3.2.2 Geometry-Induced Flow Verification (G2F, Reverse Path)

The reverse path operates in the complementary direction: starting from 3D reconstruction, it synthesizes an expected 2D motion field and evaluates its agreement with the independently estimated optical flow in *motion space*. This path covers the blind spots of the forward path—if the reconstruction model produces locally consistent but globally incorrect geometry, the resulting synthetic flow will diverge from the independently observed motion, exposing the violation.

Concretely, given the reconstructed point maps $\mathbf{P}_i$, $\mathbf{P}_j$ and the relative camera transformation $[\mathbf{R} \mid \mathbf{t}]$ estimated between the two frames from the reconstructed geometry, we compute a geometric flow field by projecting each 3D point in frame i into the coordinate system of frame j:

$$\hat{\mathbf{f}}_{i\to j}(\mathbf{p}) = \pi(\mathbf{R}\,\mathbf{P}_i(\mathbf{p}) + \mathbf{t}) - \mathbf{p}, \tag{4}$$

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where $\pi(\cdot)$ denotes perspective projection. This geometric flow represents the 2D displacement that *would* be observed if the video depicted a geometrically consistent 3D scene undergoing the estimated camera motion. The verification error measures the pixel-level ℓ_2 discrepancy between this geometry-induced prediction and the independently estimated optical flow:

$$e_{\text{flow}}(\mathbf{p}) = \left\| \hat{\mathbf{f}}_{i\to j}(\mathbf{p}) - \mathbf{f}_{i\to j}(\mathbf{p}) \right\|_2. \tag{5}$$

Large residuals indicate that the 3D structure and the observed 2D motion are mutually inconsistent, exposing geometric violations such as incorrect depth ordering, object warping, or erroneous camera pose. The path-level error is averaged analogously: $E_{\text{flow}} = \frac{1}{|\Omega|} \sum_{\mathbf{p}\in\Omega} e_{\text{flow}}(\mathbf{p})$.

3.2.3 Bidirectional Fusion

The two paths are defined in different spaces with different error magnitudes: E_{pos} measures relative 3D displacement (typical range $\sim$0.05–0.5) while E_{flow} measures pixel-level residuals (typical range $\sim$2–20). We convert each to a reward via exponential mapping with per-path temperature parameters, and fuse the two via geometric mean to ensure that *both* paths must agree for a high reward:

$$R_{\text{bgfv}} = \sqrt{R_{\text{pos}} \cdot R_{\text{flow}}}, \quad R_{\text{pos}} = \exp(-\lambda_{\text{pos}} \cdot E_{\text{pos}}), \quad R_{\text{flow}} = \exp(-\lambda_{\text{flow}} \cdot E_{\text{flow}}), \tag{6}$$

The geometric mean is a deliberate choice: it yields a high reward only when both verification paths report low error, so a geometric violation that escapes detection in one space is still penalized if the complementary path flags it. This bidirectional fusion directly addresses the fundamental limitation of unilateral evaluation identified in §1. By spanning both position space and motion space with independently sourced signals, BGFV ensures that the blind spots of one projection direction are structurally covered by the other.

3.3 Hierarchical Temporal Evaluation

A single frame-pair offset cannot capture the full spectrum of 3D violations: adjacent frames reveal local jitter and deformation, while distant frames expose cumulative drift and global structural collapse. We therefore evaluate BGFV at three temporal scales, $\delta_1 = 1$, $\delta_2 = \max(2, \lfloor N/3 \rfloor)$, $\delta_3 = \max(3, \lfloor 2N/3 \rfloor)$, targeting local smoothness, structural persistence, and global consistency respectively. At each level l, BGFV is averaged over all valid pairs with offset δ_l to yield $R^{(l)}_{\text{bgfv}}$. The levels are combined with curriculum-scheduled weights: $w_1 = 1, w_2 = \min(1,\ 2\rho), w_3 = \max(0,\ 2\rho - 1), R^{\text{hier}}_{\text{bgfv}} = \sum_{l=1}^{3} \bar{w}_l\, R^{(l)}_{\text{bgfv}}$, where $\rho = k/K$ is the training progress and $\bar{w}_l = w_l / \sum_{l'} w_{l'}$. This curriculum focuses early training on local consistency and progressively engages longer-range verification as the model stabilizes.

3.4 Dataset Curation

We curate a dedicated training set of 2k scenes from DL3DV [39], GTA-SFM [40], and MVS-Synth [41], collectively covering diverse indoor, outdoor, urban, and natural environments with accurate camera annotations. We employ Gemini 2.5 Pro [42] to generate high-quality text captions for each scene. During the training process, we adopt the complex camera poses from WorldCompass [33], which combines composite rotations and translations of varying difficulty to expose the model to challenging camera trajectories. More details are provided in Appendix C.1.

4 Experiments

4.1 Experimental Setup

Datasets. We build our evaluation benchmark on two datasets:(1) **DL3DV** [39], a real-world multi-view dataset featuring diverse indoor and outdoor scenes. Notably, the selected scenes are disjoint from those used during training, ensuring an unbiased evaluation; and (2) **MIND** [43], a synthetic open-domain benchmark built with Unreal Engine 5, covering eight scene categories at 1080p resolution, designed to stress-test 3D consistency under complex geometry and frequent occlusions. We construct image-caption pairs from 300 scenes of DL3DV and 250 scenes of MIND, and design 8 types of complex camera trajectories as input poses for evaluation. Following §3.2, we utilize Gemini 2.5 Pro [42] to generate the captions. More details are provided in Appendix C.2.

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Table 1: Quantitative comparison on the **DL3DV** dataset [39]. **Best** results are highlighted in darker blue, and **second best** in lighter blue.

Method	3D Consistency					Visual Quality	
	Reproj.↓	Epipolar↓	Bi err.↓	Flow↓	Point↓	HPSv3↑	T-SSIM↑
DiffusionAsShader [14]	0.0079	5.5012	2.4534	20.5061	0.4704	1.9719	0.6761
Trajectory-Attention [19]	0.0051	7.0090	1.0776	9.8631	0.1901	0.2217	0.7512
WorldPlay [3]	0.0022	2.8218	0.8715	10.2976	0.1585	2.4161	0.7193
w/ VideoGPA [11]	0.0025	1.9771	0.6178	4.8464	0.1311	3.7980	0.7594
w/ Epipolar error [12]	0.0019	2.0517	0.6283	4.5241	0.1993	3.4526	0.7218
w/ Depth reproj. [44]	0.0022	2.1463	0.5821	4.1173	0.1284	3.7215	0.7384
WorldCompass [33]	0.0049	2.8192	1.0067	18.2885	0.1782	3.2482	0.5938
WORLDPRISM (Ours)	0.0017	1.7158	0.5001	3.5592	0.1072	3.9554	0.7700

Table 2: Quantitative comparison on the **MIND** dataset [43]. **Best** results are highlighted in darker blue, and **second best** in lighter blue.

Method	3D Consistency					Visual Quality	
	Reproj.↓	Epipolar↓	Bi err.↓	Flow↓	Point↓	HPSv3↑	T-SSIM↑
DiffusionAsShader [14]	0.0071	5.3940	3.8087	90.0817	1.2890	0.5618	0.7466
Trajectory-Attention [19]	0.0049	6.6690	2.7520	38.0220	0.5027	0.2290	0.7999
WorldPlay [3]	0.0041	3.9612	2.0815	31.4461	0.5819	2.2986	0.7721
w/ VideoGPA [11]	0.0029	2.4350	0.9701	5.4022	0.3960	2.5700	0.8050
w/ Epipolar error [12]	0.0032	1.6799	0.9986	5.5603	0.5245	3.2527	0.8100
w/ Depth reproj. [44]	0.0037	1.5582	0.7975	4.6563	0.2791	3.6327	0.8255
WorldCompass [33]	0.0050	3.8168	1.5780	15.0367	0.4461	3.6289	0.8105
WORLDPRISM (Ours)	0.0021	1.5797	0.4626	3.0875	0.1058	3.5966	0.8513

Baselines and Metrics. We compare WORLDPRISM against two categories of baselines: (1) *Non-RL methods* that enforce geometric constraints at inference time, including DiffusionAsShader [14] and Trajectory-Attention [19]; and (2) *RL-based post-training approaches* built upon our base model WorldPlay [3], including VideoGPA [11] (a 2D perceptual evaluation reward), epipolar error [12] (a single geometric constraint), and depth reprojection error [44] (a unilateral depth-based reward). We additionally compare with WorldCompass [33], a concurrent RL-based method that focuses on 2D appearance consistency. For fair comparison, all RL-based variants adopt DiffusionNFT [31] as the optimization algorithm. We evaluate generated videos along two complementary axes: (1) *3D Consistency*, measured by reprojection error (Reproj.), epipolar error (Epipolar), bidirectional error (Bi err.), optical flow error (Flow), and point correspondence error (Point). (2) *Visual Quality*, assessed via human preference score (HPSv3) [45] and temporal structural similarity (T-SSIM).

Implementation Details. We build WORLDPRISM on top of WorldPlay, a flow-matching diffusion transformer operating in latent space. During post-training, we generate $G = 12$ candidate rollouts per training sample using the EMA policy and select the top-M and bottom-M ($M = 3$) samples for contrastive learning via DiffusionNFT [31]. The reward weights are set to $\alpha_{\text{bgfv}} = 1.0$, $\alpha_{\text{act}} = 1.0$, $\alpha_{\text{drift}} = 2.0$. The temperature parameters for bilateral fusion are $\lambda_{\text{pos}} = 5.0$ and $\lambda_{\text{flow}} = 0.5$. We use UFM [30] for optical flow estimation and Pi3 [28] for 3D point map reconstruction. Training is conducted on $16\times$ GPUs for 2 days. The hierarchical temporal evaluation uses offsets $\delta_1 = 1$, $\delta_2 = \lfloor N/3 \rfloor$, $\delta_3 = \lfloor 2N/3 \rfloor$ with curriculum scheduling over the full training horizon.

4.2 Main Results

Quantitative Results. Tab. 1 and 2 report the main quantitative results on the DL3DV [39] and MIND [43] datasets. Equipped with our proposed Bilateral Cross-Space Verification, WORLDPRISM consistently outperforms a broad spectrum of baselines across almost all 3D consistency metrics, including (i) architecture-level approaches that inject geometric priors into the generator (DiffusionAsShader [14]), (ii) camera-conditioned methods relying on explicit trajectory control

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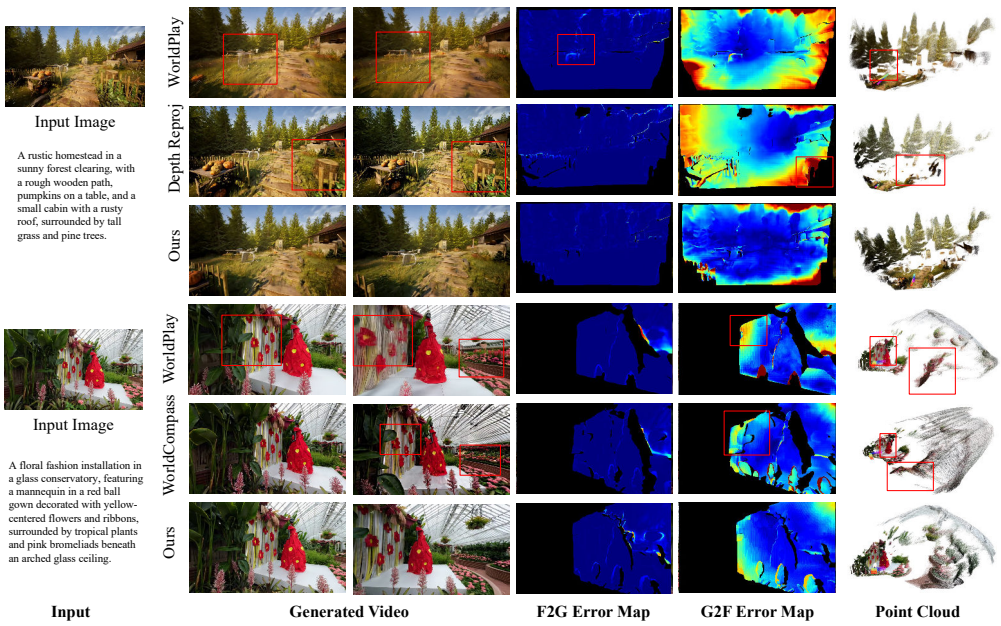


Figure 4: **Qualitative Comparison.** Key regions are highlighted with red bounding boxes. F2G Error Map measures the MSE of 3D point positions mapped via pseudo optical flow and reprojected back to the source frame. G2F Error Map measures the MSE error between the synthetic optical flow and the pseudo optical flow for each pixel on the source frame. Warmer colors indicate larger errors. **Note** that F2G errors are dominated by a few high-value outliers and most regions appear dark, whereas G2F errors have a narrower value range, yielding a brighter map.

(Trajectory-Attention [19]), (iii) post-training methods explicitly targeting 3D consistency [11, 12, 44], and (iv) post-training methods optimizing only 2D appearance, such as WorldCompass [33].

Beyond the metrics directly aligned with our reward (Flow and Point), WORLDPRISM also delivers strong gains on metrics that are *not* part of the optimization objective, including Reproj., Epipolar, and Bi err. This indicates that bilateral verification captures a more general notion of 3D consistency, of which existing unilateral objectives can be viewed as special cases. Crucially, these geometric improvements do not trade off against perceptual quality: WORLDPRISM attains the best T-SSIM on both datasets and the best HPSv3 on DL3DV while remaining competitive on MIND, confirming that the auxiliary reward terms effectively prevent quality regression during post-training.

Qualitative Results. Fig. 4 shows visual comparisons across representative scenes. In the first scene (rustic homestead), WorldPlay fails to preserve object permanence: the white object inside the highlighted region disappears in subsequent frames, indicating a violation of temporal and geometric consistency. In addition, the surrounding grass exhibits severe blurring, reflecting instability in fine-grained structures. For Depth Reprojection, the wooden stump in the highlighted region shows significant spatial drift in 3D space. Although this misalignment is evident in the reconstructed geometry, it is less noticeable in 2D image space, revealing the limitation of single-view projection-based supervision. In contrast, our method preserves both object presence and spatial alignment by jointly enforcing consistency in both 2D and 3D spaces through complementary reward signals. In the second scene (glass conservatory), WorldPlay again produces noticeable blurring on the flowers attached to the wall, indicating a lack of structural stability. Depth Reprojection suffers from clear geometric distortion on the red brick wall behind the flower bed. WORLDPRISM mitigates these issues by enforcing cross-space consistency, leading to sharper object boundaries and more coherent geometric layouts across frames. More visualization results are provided in Appendix D.

4.3 Ablation Study

To further validate the effectiveness of each core component, we conduct an ablation study on the MIND [43] dataset, as shown in Tab. 3.

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Table 3: Ablation study on the MIND dataset [43]. Each row removes one component from the full WORLDPRISM model. **Best** results are highlighted in darker blue, and **second best** in lighter blue.

Method	3D Consistency					Visual Quality	
	Reproj.↓	Epipolar↓	Bi err.↓	Flow↓	Point↓	HPSv3↑	T-SSIM↑
WORLDPRISM (Full)	0.0021	1.5797	0.4626	3.0875	0.1058	3.5966	0.8513
w/o F2G	0.0021	2.0509	0.7211	5.4574	0.1889	2.9173	0.8435
w/o G2F	0.0023	2.8806	0.8789	7.7379	0.1577	1.7200	0.7773
w/o Scale-Invariant	0.0023	2.2879	0.7531	6.7170	0.1327	2.5656	0.7973
w/o Hierarchical Temp.	0.0020	2.1544	0.8100	6.0527	0.2102	2.9556	0.8135

(1) Flow-Guided Geometry Verification (Forward Path). Removing the forward path leads to a clear degradation in 3D geometric accuracy, particularly in point-level and motion consistency. The point error increases from 0.1058 to 0.1889 (+78.5%), while the flow error rises from 3.0875 to 5.4574 (+76.7%). This substantial deterioration indicates that lifting 2D correspondences into 3D space is a critical component of our framework. It effectively constrains depth ordering and reduces lateral drift that cannot be reliably resolved in the 2D domain alone.

(2) Geometry-Induced Flow Verification (Reverse Path). Removing the reverse path also results in consistent performance degradation across multiple metrics. The epipolar error increases from 1.5797 to 2.8806, while the flow error increases from 3.0875 to 7.7379. In addition, visual quality HPSv3 decreases from 3.5966 to 1.7200. These results suggest that the reverse path plays a complementary role to the forward path by identifying globally inconsistent yet locally plausible 3D structures.

(3) Scale-Invariant Formulation. Replacing the scale-invariant formulation with direct Euclidean comparison degrades epipolar error from 1.58 to 2.29 and flow error from 3.09 to 6.72. This indicates that monocular scale ambiguity introduces systematic noise if not properly normalized.

(4) Hierarchical Temporal Evaluation. Using only adjacent-frame evaluation increases the point error (0.11→0.21) and increases bidirectional error (0.46→0.81). While single-offset evaluation captures local jitter, it fails to detect long-term drift, which is revealed by hierarchical multi-scale offsets.

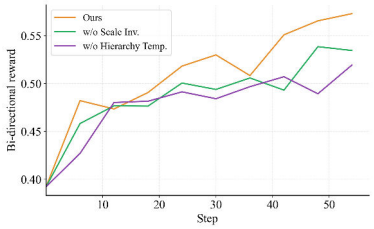


Figure 5: **Reward score curves during training.**

To further prove the effectiveness of our bidirectional design, we visualize the 3D consistency error maps under both two paths. As shown in Fig. 2, either single path fails to detect the 3D inconsistencies: (1) The flow-based path (G2F) fails to detect structural distortions; (2) The geometry-verified path (F2G) fails to detect visual artifacts. In contrast, our reward leverages complementary signals from both paths to successfully detect 3D inconsistencies in the generated video. The reward curve during the training process, shown in Fig. 5, further shows the effectiveness of our proposed Scale-Invariant Formulation and Hierarchical Temporal Evaluation.

5 Conclusion

We presented WORLDPRISM, a post-training framework that improves the 3D consistency of video world models via Bidirectional Geometry-Flow Verification, which cross-checks optical flow and 3D reconstruction through two complementary paths (F2G and G2F) so that the blind spots of each single evaluation space are structurally covered by the other. Combined with scale-invariant metrics and hierarchical temporal evaluation, our reward provides a robust 3D consistency signal that avoids the systematic limitations of unilateral 2D-based approaches. We build our WORLDPRISM with World-Play as the base model, and experiments on diverse benchmarks show that WORLDPRISM achieves state-of-the-art 3D consistency while preserving visual quality and generative diversity. We present bidirectional cross-space verification as a preliminary step toward more robust reward design in video generation, and we hope that the proposed WORLDPRISM serves as a useful reference for future efforts in building 3D-consistent video world models.

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A Additional Quantitative Results

To further validate that our post-training procedure does not compromise general video generation capabilities, we evaluate WorldPrism and baselines on VBench [46], a comprehensive benchmark suite covering diverse aspects of video quality beyond 3D consistency. As shown in Tab. 4, WorldPrism achieves the best or second-best performance across nearly all dimensions. Compared to our base model WorldPlay [3], WorldPrism yields substantial improvements on consistency-related metrics (Subject Consistency: $0.9104 \rightarrow 0.9483$, Background Consistency: $0.9117 \rightarrow 0.9255$) while simultaneously improving visual quality indicators such as Aesthetic Quality ($0.4791 \rightarrow 0.4926$) and I2V fidelity. This confirms that the auxiliary reward terms (R_{act} and R_{drift}) effectively prevent quality degradation during geometric-focused post-training. Compared to WorldCompass [33], WorldPrism delivers notably stronger Subject Consistency (0.9483 vs 0.8969) and Background Consistency (0.9255 vs 0.8991), showing that enforcing explicit 3D geometric constraints translates to more coherent temporal consistency as perceived by VBench's evaluation protocols. These results indicate that our Bidirectional Geometry-Flow Verification steers the model toward genuine 3D-consistent generation rather than superficial appearance matching, and that such geometric grounding benefits overall video quality rather than trading off against it.

Table 4: Evaluation on public benchmark VBench [46].

Method	I2V Subject	I2V Background	Subject Consistency	Background Consistency	Motion Smooth	Aesthetic Quality	Imaging Quality
DiffusionAsShader [14]	0.9443	0.9618	0.8957	0.9159	0.9828	0.4764	0.6250
Trajectory-Attention [19]	0.9481	0.9622	0.9314	0.9324	0.9859	0.4699	0.6167
WorldPlay [3]	0.9736	0.9767	0.9104	0.9117	0.9916	0.4791	0.6780
WorldCompass [33]	0.9732	0.9745	0.8969	0.8991	0.9884	0.4908	0.7164
WORLDPRISM (Ours)	0.9822	0.9820	0.9483	0.9255	0.9923	0.4926	0.6896

B Details of the WORLDPRISM

B.1 Post-Training via DiffusionNFT

Reward composition. Optimizing solely for geometric consistency risks degrading action fidelity or visual quality, as the model may exploit degenerate solutions (static frames or minimal camera motion) that trivially minimize geometric error. We therefore combine the hierarchical BGFV signal with two auxiliary objectives adopted from WorldCompass [33]:

$$R = \alpha_{\text{bgfv}}\, R^{\text{hier}}_{\text{bgfv}} + \alpha_{\text{act}}\, R_{\text{act}} + \alpha_{\text{drift}}\, R_{\text{drift}}, \tag{7}$$

where R_{act} measures action-following accuracy by comparing the camera motion estimated from the generated clip against the conditioning action sequence, ensuring that the model faithfully executes the instructed camera trajectory rather than defaulting to geometrically trivial motions. R_{drift} penalizes visual quality degradation relative to the conditioning frame by measuring perceptual similarity, preventing the model from sacrificing texture fidelity, sharpness, or color consistency in pursuit of geometric accuracy. Together, these auxiliary signals form a multi-objective optimization landscape that prevents the model from exploiting geometric consistency at the expense of action fidelity or perceptual quality, maintaining a balanced improvement across all axes.

Optimization. We adopt DiffusionNFT [31], a negative-aware fine-tuning algorithm specifically designed for flow matching models. Unlike standard RLHF approaches that only reinforce high-reward samples, DiffusionNFT leverages both high-reward and low-reward samples through a contrastive learning framework, enabling more sample-efficient optimization by simultaneously pulling the model toward desirable outputs and pushing it away from undesirable ones.

Given G rollout clips $\{x_n^{(i)}\}_{i=1}^{G}$ generated by the EMA policy $\pi_{\theta_{\text{old}}}$ for a given training sample, each clip receives per-channel reward scores $\{s_j^{(i)}\}_{j\in\{\text{bgfv,act,drift}\}}$. We first compute per-channel advantages via z-score normalization to handle the different scales of reward signals:

$$a_j^{(i)} = \frac{s_j^{(i)} - \text{mean}(\{s_j^{(i)}\}_{i=1}^{G})}{\text{std}(\{s_j^{(i)}\}_{i=1}^{G})}, \tag{8}$$

which centers each reward channel to zero mean and unit variance within each batch of rollouts, removing the need for manual reward scaling. The normalized advantages are then aggregated into a single optimality weight:

$$r^{(i)} = \frac{1}{2} + \frac{1}{2}\,\text{clip}\left[\frac{\sum_j \alpha_j\, a_j^{(i)}}{Z},\ -1,\ 1\right], \tag{9}$$

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where $Z = \sum_j \alpha_j$ is a normalizing constant. The resulting $r^{(i)} \in [0, 1]$ serves as a soft interpolation coefficient: values near 1 indicate high-quality samples that should be reinforced, while values near 0 indicate low-quality samples that should be suppressed.

From the G candidates, we select the top-M and bottom-M by total reward to form a contrastive training subset $\mathcal{G}_{\text{sub}}$, discarding intermediate samples that provide weaker learning signal. For each selected sample, we construct noisy latents at randomly subsampled timesteps $t \sim \mathcal{T}_{\text{sub}}$:

$$\mathbf{z}_t^{(i)} = (1 - t)\, x_n^{(i)} + t\, \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{10}$$

and define implicit positive/negative velocity targets through an interpolation between the EMA policy and the current model:

$$v_\theta^+ = (1 - \beta)\, v_{\theta_{\text{old}}}(\mathbf{z}_t^{(i)}, t) + \beta\, v_\theta(\mathbf{z}_t^{(i)}, t), \tag{11}$$

$$v_\theta^- = (1 + \beta)\, v_{\theta_{\text{old}}}(\mathbf{z}_t^{(i)}, t) - \beta\, v_\theta(\mathbf{z}_t^{(i)}, t), \tag{12}$$

where $v_{\theta_{\text{old}}}$ and v_θ denote predictions from the EMA policy and the current model respectively, and β controls the interpolation strength. Intuitively, v_θ^+ represents a target that slightly amplifies the current model's deviation from the EMA baseline (reinforcing its predictions), while v_θ^- represents a mirror target that penalizes the current deviation (suppressing its predictions).

The final training objective combines both positive and negative supervision weighted by the optimality coefficient:

$$\mathcal{L}(\theta) = \mathbb{E}_{\substack{i \sim \mathcal{G}_{\text{sub}} \\ t \sim \mathcal{T}_{\text{sub}}}} \left[r^{(i)} \left\| v_\theta^+ - v^{(i)} \right\|_2^2 + (1 - r^{(i)}) \left\| v_\theta^- - v^{(i)} \right\|_2^2 \right], \tag{13}$$

where $v^{(i)} = x_n^{(i)} - \boldsymbol{\epsilon}$ is the ground-truth velocity target derived from the clean sample and the noise. For high-reward samples ($r^{(i)} \approx 1$), the loss is dominated by the positive term, reinforcing the current model's velocity predictions. For low-reward samples ($r^{(i)} \approx 0$), the negative term dominates, optimizing toward the mirror velocity v_θ^- and effectively steering the model away from generating low-quality outputs. This asymmetric contrastive formulation enables the model to learn from both success and failure modes simultaneously, leading to more stable and efficient convergence compared to purely positive reinforcement approaches.

C Experimental Settings

C.1 Details of Training Dataset Curation

Our training set is curated from three complementary sources that collectively provide diverse real-world and synthetic scenes, totaling 2,003 video sequences.

DL3DV [39]. DL3DV-10K is a large-scale real-world scene dataset containing over 10,000 videos captured at 4K resolution, covering 65 diverse categories including both indoor and outdoor environments (e.g., offices, streets, museums, parks). We select 1,000 representative scenes from this dataset (49.9% of our training set), prioritizing diversity in scene layout, scale, and appearance. These real-world sequences provide naturalistic camera trajectories and complex visual phenomena such as reflections, transparency, and repetitive textures.

GTA-SFM [40]. GTA-SFM is a synthetic dataset rendered from the Grand Theft Auto V (GTA V) game engine, featuring a variety of urban driving scenarios with diverse environmental conditions including different weather, time of day, and lighting. We utilize 643 sequences from this dataset (32.1%), which provide geometrically consistent multi-view imagery under controlled settings that complement the real-world data.

MVS-Synth [41]. MVS-Synth is another synthetic dataset rendered from the GTA V engine, specifically designed for multi-view stereo research. It contains 120 urban scenes, each comprising 100 high-resolution RGB frames (1920×1080), covering challenging visual conditions such as textureless surfaces, reflective materials, and thin structures. We include 360 sequences from this dataset (18.0%), leveraging its diverse urban environments to improve generalization in scenarios where real-world data coverage is limited.

C.2 Details of Benchmark Setup

Our evaluation benchmark comprises 550 scenes from two sources: 300 scenes from DL3DV [39] (disjoint from the training split) and 250 scenes from MIND [43]. For each scene, we design 16 types of camera trajectories covering both simple single-axis motions and complex compound motions, as summarized in Table 5.

These trajectories range from single-axis translations/rotations (ID 0–3) to round-trip motions (ID 4–7), combined motions (ID 8–9), and complex multi-axis trajectories (ID 10–15), enabling fine-grained diagnosis of 3D consistency under increasing motion complexity.

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Table 5: The 16 camera trajectory types used in our evaluation benchmark.

ID	Trajectory	ID	Trajectory
0	Move Forward	8	Forward + Turn Left
1	Move Left	9	Forward + Look Up
2	Turn Left	10	Orbit Left
3	Look Up	11	Zigzag
4	Forward-Backward	12	U-Turn
5	Left-Right	13	Pan 360
6	Turn Left-Right	14	Crane Up
7	Look Up-Down	15	Spiral

D Additional Qualitative Results

We provide additional qualitative results in Fig. 6–9 to further demonstrate the effectiveness of our proposed WORLDPRISM in improving the 3D consistency of videos generated by the HunyuanVideo model [4] through post-training. For each comparison, we visualize the generated video frames, the F2G error maps, the G2F error maps, and the reconstructed 3D point clouds.

As shown in Fig. 6 and Fig. 7, the baseline methods (WorldPlay [3], VideoGPA [11], and Depth Reprojection from VGGRPO [44]) exhibit noticeably warmer colors in both the F2G and G2F error maps, indicating larger geometric inconsistencies in the generated videos. In contrast, WORLDPRISM produces substantially lower error across both verification paths, confirming that the bilateral cross-space reward effectively enforces mutual agreement between 2D motion and 3D structure. The reconstructed point clouds further corroborate these observations: baselines produce fragmented or distorted geometry, whereas WORLDPRISM yields coherent and structurally plausible 3D reconstructions.

Fig. 8 and Fig. 9 present direct frame-level comparisons across diverse scenes. Compared with WorldPlay [3], VideoGPA [11], and Depth Reprojection from VGGRPO [44], videos generated by WORLDPRISM exhibit fewer structural distortions, reduced spatial drift, and better-preserved object boundaries under complex camera motions. These improvements are consistent across both indoor and outdoor scenarios, demonstrating that the bidirectional design of BGFV generalizes well to a wide variety of scene types and camera trajectories without sacrificing visual quality.

E Limitation

Our method is primarily designed for static scenes where all observed optical flow is induced by camera motion. When the generated video contains dynamic objects (e.g., moving pedestrians or vehicles), the current framework cannot distinguish whether the observed flow originates from camera ego-motion or independent object motion. This ambiguity leads to inaccurate reward signals: the Bidirectional Geometry-Flow Verification may incorrectly penalize valid dynamic regions or, conversely, fail to detect genuine 3D inconsistencies masked by object motion. As a result, the reward may become noisy or misleading during post-training on scenes with significant dynamic content. A promising future direction is to incorporate motion segmentation to decompose the scene into static background and dynamic foreground, enabling separate consistency evaluation for each component.

F Broader Impacts

On the positive side, improving 3D consistency in video world models can benefit robotics and autonomous driving by providing more geometrically faithful simulators, reducing the need for costly real-world data collection and enabling safer virtual evaluation. On the negative side, advances in realistic video generation could potentially be misused to create more convincing synthetic media for disinformation purposes. However, our method focuses on action-conditioned world models designed for embodied simulation rather than general-purpose video synthesis, which limits direct misuse potential. We encourage the community to develop watermarking and detection methods alongside advances in video generation.

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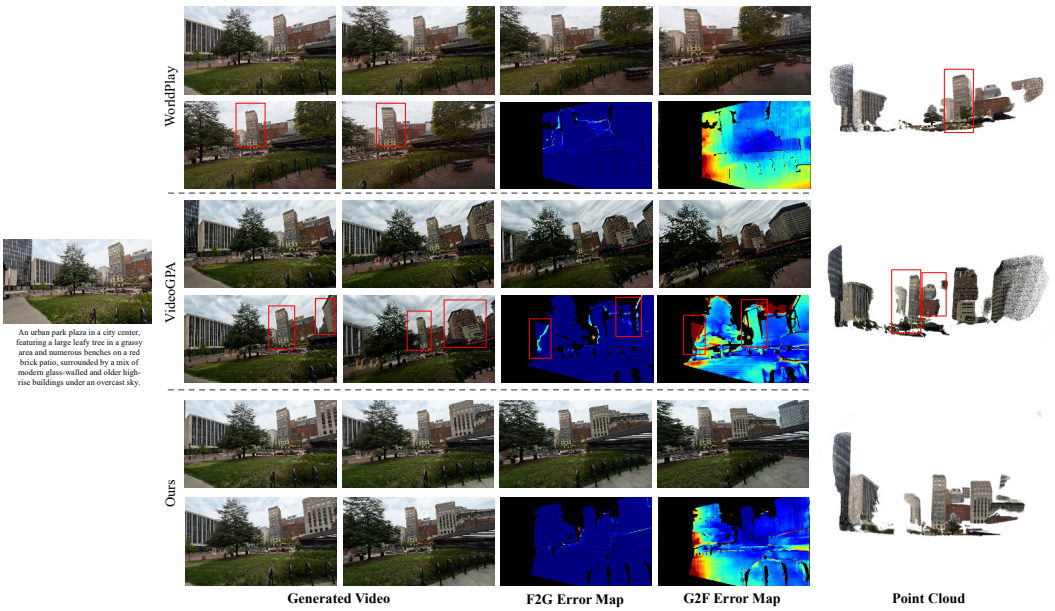


Figure 6: **Qualitative Comparison with VideoGPA and WorldPlay.** We show the generated video, F2E Error Map, G2F Error Map and the reconstructed point cloud.

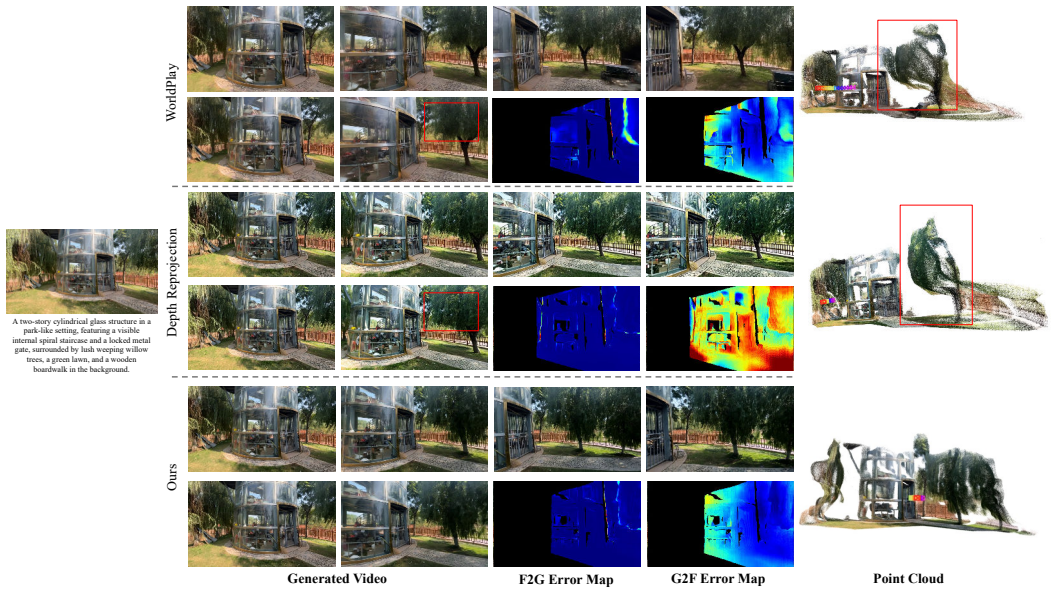


Figure 7: **Qualitative Comparison with Depth Reprojection from VGGRPO and WorldPlay.** We show the generated video, F2E Error Map, G2F Error Map and the reconstructed point cloud.

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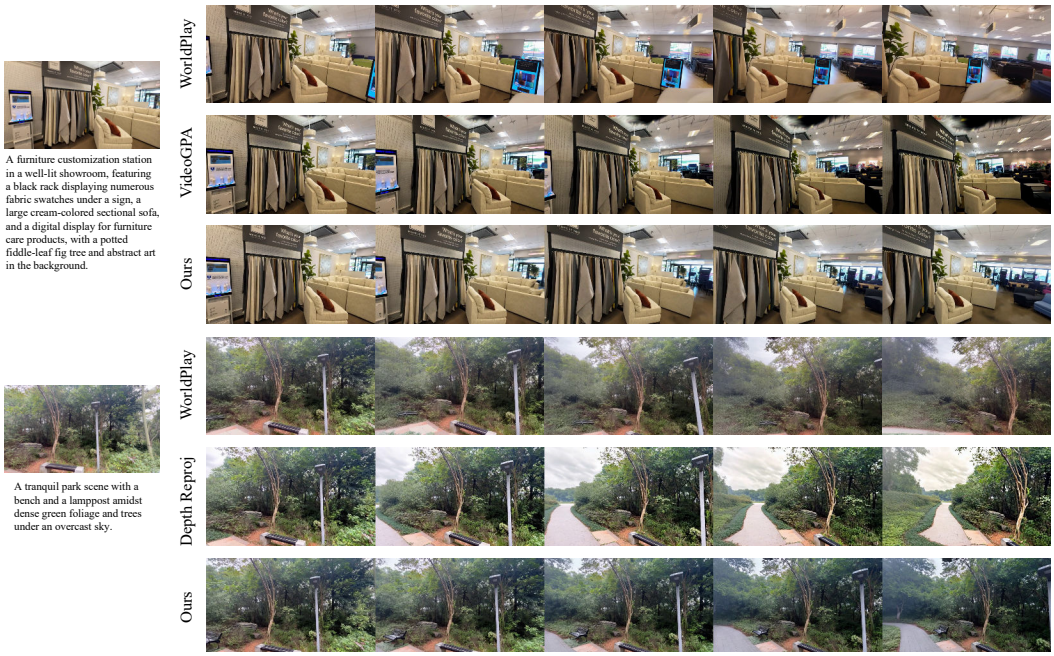


Figure 8: **Visualization of the generated videos.** Compared with other baseline models (WorldPlay, VideoGPA, and Depth Reprojection from VGGRPO), we find that the generated videos by our proposed WORLDPRISM have fewer structure distortions and artifacts, showing better 3D consistency.

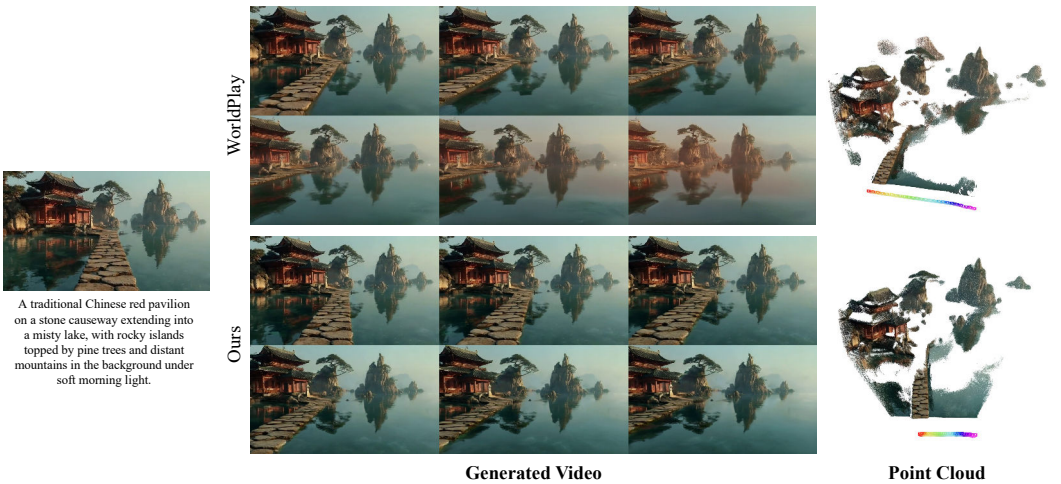


Figure 9: **Visualization of the generated videos.** We show the generated video and the reconstructed point cloud compared with WorldPlay, where our proposed method achieves 3D-consistent world modeling.

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